Taiwo Alabi

Data Scientist. I build AI systems end-to-end and publish exactly where each one fails

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Profile

I tune intelligence to the cost of being wrong. Every model has a dial, and the setting depends on the actual weight of a mistake in real life. I learned that on a telecom platform carrying 400 million usage records a day, where a bad deploy meant fourteen downstream systems read nothing, or read twice. I am a Master's student in Artificial Intelligence with five systems designed and produced end to end, one deployed and publicly reachable, and I can explain every one of them to you during our interview.

Professional Experience

Turing, Remote (Dublin) Apr 2026 to May 2026
AI Quality Analyst, Google Gemini evaluation programme

- I ***measured*** the quality of ***Google Gemini*** responses on ***personalization*** features, then wrote prompts to test the models, and documented the process for a ***Reinforcement Learning from Human Feedback (RLHF)*** project.

Huawei Technologies (via RGS), Globacom Telecommunications department, Lagos Summer 2024 & 2025
Product Manager Intern

- Problem: On a national mobile network in one of ***Africa's largest mobile markets, 400 million daily call records represent 400 million bills*** that real people pay.
- Action: I designed and tested the API layer for a cloud integration between the contact centre platform and a commercial bank's customer care line. I tracked every interface the bank had to expose against what we had actually received, and tested each one with curl before it went near a voice prompt.
- Result: When authentication fails, the call does not retry silently and does not dead-end. **It transfers to a person.** That is the rule I have built into every system since.
- I also worked with the solutions architect on a data migration that partitions customer data into domain-specific buckets, and with the test team designed over 440 User Acceptance Test (UAT) cases. During the phased cutover a downstream storage target filled and blocked uploads, and a control connection reset stopped one CDR stream. Both surfaced in verification, before any customer saw them.

Selected Projects

Smart Medical Records Engine FastAPI, PostgreSQL/pgvector, LangGraph, Docker

- **Problem:** A doctor has ***ten minutes*** and a patient history spread across ***years of notes***, the one critical line, ***a past reaction to a medicine***, gets missed.
- **Action:** I built a ***clinical assistant*** that answers ***plain-English questions*** from that patient's own records, with retrieval ***scoped per patient*** so no other patient's notes can surface, plus ***physician authorization***, ***PII redaction***, and an ***audit log of every access***.
- **Result:** I ran the ***whole stack***, the API, ***PostgreSQL with 768-dimensional vector search*** and a ***self-hosted Llama 3.1***, across ***four Docker Compose services***, so ***patient data never leaves the hospital's own network***.
  Read the write-up (PDF) | **Architecture diagram** | **Data model** | **Source code**

Customer Care Process Automation Python, scikit-learn, BPMN 2.0

- **Problem:** Hundreds of customer complaints used to get ***sorted by hand every morning***, which means we used to have more misrouted tickets because of ***human errors***. Customers ***waited for minutes*** until proper routing directed their ticket to the right back-end team, followed by a few minutes of communication with that customer to resolve the issue.
- Action: I modeled the process in Business Process Model and Notation first, then split it: a rule-based bot enforcing field-level conformance against a standard report catalogue, the kind of signed field definitions I worked to at Huawei, and an AI agent reading customer complaint tickets and routing them to the appropriate back-end team. It reached 95.2% routing accuracy on held-out tickets, 40 of 42, a corpus small enough that I report the count and not just the rate.

- **Result:** I built a ***confidence gate*** so the agent ***escalates uncertain tickets to a human*** instead of guessing. On ***4,000 unseen tickets***, it correctly labeled every ticket it routed and appropriately flagged the uncertain edge cases for human review.

  Read the write-up (PDF) | BPMN process models | Confusion matrix | Source code

TermsGuard AI Sentence transformers, FAISS, Claude API, Flask

- Problem: A 2008 study estimated that reading every privacy policy an average internet user meets would take about 244 hours a year. A 2017 Deloitte survey of 2,000 consumers found 91% accept terms without reading them.
- **Action:** I contributed to a ***retrieval-augmented pipeline*** that flags ***unfair clauses in plain English***, reaching 87% F1 and cutting ***model hallucination by 77%*** through grounding output in retrieved reference clauses.
- **Results:** I cut ***end-to-end latency from 28 seconds to under 10***, and added a ***Claude verification stage*** that removes false positives, and built a cosine-only fallback when no API key is set.

  Live demo | Read the write-up (PDF) | Source code

Automated QA Gate for AI Generated Imagery TensorFlow/Keras, ResNet50, scikit-learn

- **Problem:** ***AI-generated fashion campaign images*** silently drift over multiple prompt generations; the character's ***identity starts to shift*** or specific details I want to retain become inconsistent**.**
- Action: I built a two-stage gate: a Siamese network on a frozen ResNet50 to flag identity drift and an autoencoder for style drift, reaching precision 1.00 with zero false positives on a 90-pair test set, against pixel difference and kNN baselines at F1 0.59 and 0.26. Tuning the decision threshold from the sigmoid default of 0.5 to 0.4821, on validation data only, recovered all eight missed pairs without adding a single false positive.

  Read the write-up (PDF) | ROC curve and confusion matrix | Architecture diagram | Source code

Education

National College of Ireland, Dublin Jan 2026 to Jan 2027
MSc Artificial Intelligence (part time)

Modules: Machine Learning, ***Intelligent Agents and Process Automation, AI-Driven Decision Making, Engineering and Evaluating AI, Natural Language Processing, Data Governance and Ethics,*** Database Programming, Foundations of AI.

Afe Babalola University, Lagos Sep 2021 to Feb 2025
BSc Computer Science, Second Class Honours, Upper Division (2.1)

Technical Skills

Languages: Python, SQL

Data & Infrastructure: Hadoop HDFS, HBase, GaussDB, PostgreSQL, pgvector, Docker, Docker Compose, Kubernetes, FastAPI, Flask, Google Cloud

Machine Learning: scikit-learn, pandas, TensorFlow/Keras, transfer learning, reinforcement learning, Q-Learning, neural networks, Transformersgenetic algorithms

AI & Agents: RAG pipelines, vector search (pgvector, FAISS), LangGraph, LLM evaluation, genetic algorithms

Data Governance: GDPR data cataloguing, data lineage, data quality validation, UAT design

Tools: Git, GitHub, Claude Code, Linux